

1. Let M be an arbitrary Turing machine. Prove the following claims.

- (a) There is a Turing machine M^R such that

$$\text{ACCEPT}(M^R) = \text{REJECT}(M)$$

and

$$\text{REJECT}(M^R) = \text{ACCEPT}(M).$$

- (b) There is a Turing machine M^A such that

$$\text{ACCEPT}(M^A) = \text{ACCEPT}(M)$$

and

$$\text{REJECT}(M^A) = \emptyset.$$

- (c) There is a Turing machine M^H such that

$$\text{ACCEPT}(M^H) = \text{HALT}(M)$$

and

$$\text{REJECT}(M^H) = \emptyset.$$

2. Let $\text{SELFACCEPT} = \{\langle M \rangle : M \text{ accepts } \langle M \rangle\}$. Prove that SELFACCEPT is undecidable.